

# Phase-controlled six-port single-photon scattering in a two-giant-atom three-waveguide node

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We analyze single-photon scattering in a waveguide-QED node formed by two giant atoms coupled to three one-dimensional waveguides. The two atoms connect three waveguides, so a photon entering from either side of any waveguide can leave through six output ports. Using a real-space scattering formulation, we derive the single-photon amplitudes by solving the full set of boundary and atomic equations. The resulting  $6 \times 6$  scattering matrix is built from six independently solved incident columns and is unitary to numerical precision. Local coupling phases provide independent interference knobs that are absent in the corresponding small-atom node; by tuning them, selected output channels can be enhanced while the other five are suppressed. The phase-reversed device is the time-reversal partner of the original one, giving the probability relation  $T_{ij}(\theta) = T_{ji}(-\theta)$  rather than the ordinary reciprocity condition  $S = S^T$ . Representative fixed-backbone parameter sets show near-unit single-photon routing probabilities and extended high-fidelity basins in phase space.

## I. INTRODUCTION

The interaction between light and matter confined to one-dimensional continua is central to waveguide quantum electrodynamics (QED) [1, 2]. When an artificial atom couples to a waveguide at multiple spatially separated points—the giant-atom regime [3, 4]—propagation phases between coupling points become controllable scattering parameters. These phase degrees of freedom enable effects without a direct small-atom analog, including frequency-dependent relaxation and Lamb shifts, decoherence-free atomic states, and non-Markovian scattering dynamics [5, 6].

For quantum-network applications, a central challenge is to route single photons among multiple output ports with high fidelity [7]. Conventional small-atom routers often rely on chiral couplings or external driving to break reciprocity. Giant-atom structures offer a complementary route by encoding routing instructions in the interference between spatially separated coupling channels. Recent work on phase-controlled giant-molecule scattering has demonstrated that local coupling phases alone can realize targeted routing and asymmetric single-photon scattering without requiring chiral couplings [8].

Here we study a two-giant-atom, three-waveguide node: atom  $a$  couples to waveguides 1 and 2, atom  $b$  couples to waveguides 2 and 3, and waveguide 2 acts as a shared bus (Fig. 1). The six external ports—left and right directions on each waveguide—make this the minimal geometry in which a single photon can be flexibly routed among all three waveguides in both directions. The system has two physically distinct incident configurations: edge incidence from waveguide 1 (or, by symmetry, waveguide 3), and middle incidence from the shared waveguide 2. Within each configuration, the routing problem reduces to selecting among six available output channels.

Our aim is to derive the scattering amplitudes in closed analytical form and expose the algebraic structure behind the routing behavior. We find that all scatter-

ing amplitudes—for both edge and middle incidence—share a common denominator  $M$  whose pole structure encodes the collective two-atom resonance. The amplitudes admit a compact  $\eta$ -form representation: each output is expressed through the atomic excitation amplitudes  $\eta_a, \eta_b$  and elementary phase factors. Three amplitudes ( $r_1, t_1, r_2$ ) further reduce to factorized rational forms, while the remainder retain their  $\eta$ -form structure. Many routing conditions reduce to simple phase-factor zeros, while the waveguide-2 outputs are governed by linear-combination zeros, providing a compact classification of representative routing mechanisms.

The paper is organized as follows. Section II introduces the model, Hamiltonian, and real-space scattering formalism, including the edge-incidence factorized amplitudes and the zero-condition covering classification. Section III visualizes the edge-incidence routing landscape under the fixed backbone  $\Gamma = 1, \theta_{a2} = \pi$ . Section IV derives the middle-incidence amplitudes and uses them, together with the edge-incidence results and independently derived right-incidence columns, to construct the complete  $6 \times 6$  scattering matrix. Section V discusses finite-delay extensions and directions for future work.

## II. MODEL AND HAMILTONIAN

### A. Model configuration

The system consists of three infinite one-dimensional waveguides and two two-level giant atoms. Atom  $a$  couples to waveguide 1 at two points ( $x = 0$  and  $x = l_1$ ) and to waveguide 2 at two points ( $x = 0$  and  $x = l_2$ ). Atom  $b$  couples to waveguide 2 at the same two points as atom  $a$ , and to waveguide 3 at two points ( $x = 0$  and  $x = l_3$ ). The shared bus waveguide 2 thus mediates an effective interaction between the two atoms.

The six external ports are labeled by the asymptotic

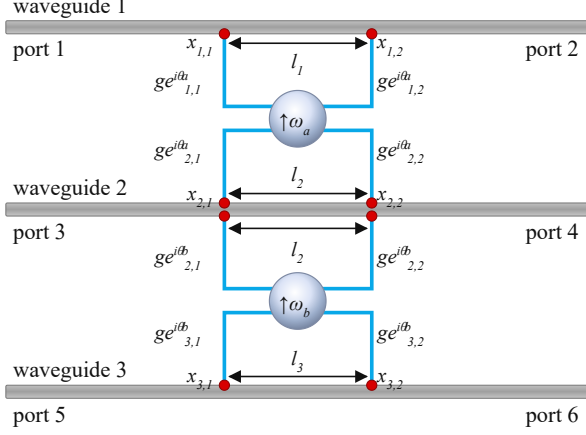


FIG. 1. Two-giant-atom, three-waveguide six-port node. Atom  $a$  couples to waveguides 1 and 2, while atom  $b$  couples to waveguides 2 and 3. For WG1-L incidence, the output amplitudes are  $r_1$  (port 1),  $t_1$  (port 2),  $r_2$  (port 3),  $t_2$  (port 4),  $r_3$  (port 5), and  $t_3$  (port 6).

propagation direction on each waveguide:

$$(1, 2, 3, 4, 5, 6) = (\text{WG1-L}, \text{WG1-R}, \text{WG2-L}, \text{WG2-R}, \text{WG3-L}, \text{WG3-R}), \quad (1)$$

where “L” and “R” refer to the left- and right-moving output directions, respectively. With this convention, for WG1 left incidence the reflection amplitude  $r_1$  exits through port 1 (WG1-L, left-moving) and the transmission amplitude  $t_1$  exits through port 2 (WG1-R, right-moving). The complete column vector for WG1-L incidence is therefore  $\mathbf{s}^{(1)} = (r_1, t_1, r_2, t_2, r_3, t_3)^T$ .

## B. Hamiltonian and parameters

In the rotating-wave approximation with linearized dispersion, the real-space Hamiltonian reads ( $\hbar = 1$ )

$$H = H_{\text{at}} + H_{\text{wg}} + H_{\text{int}}, \quad (2a)$$

$$H_{\text{at}} = \sum_{m=a,b} \omega_m \sigma_m^+ \sigma_m^-, \quad (2b)$$

$$H_{\text{wg}} = i v_g \sum_{j=1}^3 \int dx [L_j^\dagger(x) \partial_x L_j(x) - R_j^\dagger(x) \partial_x R_j(x)], \quad (2c)$$

$$H_{\text{int}} = \sum_{j=1,2} \int dx G_j^a(x) \sigma_a^+ [L_j(x) + R_j(x)] + \sum_{j=2,3} \int dx G_j^b(x) \sigma_b^+ [L_j(x) + R_j(x)] + \text{H.c.}, \quad (2d)$$

$$G_j^m(x) = g [e^{i\theta_{j,1}^m} \delta(x) + e^{i\theta_{j,2}^m} \delta(x - l_j)]. \quad (2e)$$

Here  $R_j^\dagger(x)$  and  $L_j^\dagger(x)$  create right- and left-moving photons in waveguide  $j$ ,  $v_g$  is the group velocity, and  $\sigma_m^\pm$  are the raising and lowering operators for atom  $m$ . The coupling function  $G_j^m(x)$  contains two local phases  $\theta_{j,1}^m$  and  $\theta_{j,2}^m$  at the two coupling points. Only their difference

$$\theta_{mj} = \theta_{j,2}^m - \theta_{j,1}^m \quad (3)$$

enters the final scattering coefficients; we fix the gauge  $\theta_{j,1}^m = 0$  throughout. The four independent phase differences are  $\theta_{a1}, \theta_{a2}, \theta_{b2}, \theta_{b3}$ . The propagation phase accumulated between the two coupling points on waveguide  $j$  is

$$\varphi_j = k_j l_j. \quad (4)$$

We define the photon-atom detuning  $\Delta \equiv \omega - \omega_a = \omega - \omega_b$  (the two atoms are taken to be degenerate), the waveguide-induced decay rate  $\Gamma \equiv g^2/v_g$ , and the normalized atomic excitation amplitudes  $\eta_m = g f_m/v_g$  ( $m = a, b$ ). It is also convenient to introduce the effective coupling parameter

$$\tau_{mj} = 2\Gamma e^{i\varphi_j} \cos \theta_{mj}, \quad (5)$$

which combines the propagation phase and the local phase difference into a single complex quantity.

The single-excitation eigenstate of the total Hamiltonian takes the standard form

$$|\Psi\rangle = \sum_{j=1}^3 \int dx [\Phi_{R,j}(x) R_j^\dagger(x) + \Phi_{L,j}(x) L_j^\dagger(x)] |\emptyset\rangle + f_a \sigma_a^+ |\emptyset\rangle + f_b \sigma_b^+ |\emptyset\rangle, \quad (6)$$

where  $|\emptyset\rangle$  is the vacuum state of the waveguide fields and both atoms are in their ground states. The photonic wavefunctions  $\Phi_{R,j}(x)$  and  $\Phi_{L,j}(x)$  are determined by the scattering boundary conditions together with the jump conditions at the coupling points, as detailed below.

### C. Scattering formulation: edge incidence

The node supports two physically distinct incident configurations. We first consider a photon incident from the left side of waveguide 1 (port 1). The photonic wavefunctions take the piecewise form

$$\Phi_{R,1}(x) = e^{ik_1x} [\Theta(-x) + A\Theta(x)\Theta(l_1 - x) + t_1\Theta(x - l_1)], \quad (7a)$$

$$\Phi_{L,1}(x) = e^{-ik_1x} [r_1\Theta(-x) + B\Theta(x)\Theta(l_1 - x)], \quad (7b)$$

$$\Phi_{R,2}(x) = e^{ik_2x} [C\Theta(x)\Theta(l_2 - x) + t_2\Theta(x - l_2)], \quad (7c)$$

$$\Phi_{L,2}(x) = e^{-ik_2x} [r_2\Theta(-x) + D\Theta(x)\Theta(l_2 - x)], \quad (7d)$$

$$\Phi_{R,3}(x) = e^{ik_3x} [E\Theta(x)\Theta(l_3 - x) + t_3\Theta(x - l_3)], \quad (7e)$$

$$\Phi_{L,3}(x) = e^{-ik_3x} [r_3\Theta(-x) + F\Theta(x)\Theta(l_3 - x)], \quad (7f)$$

where  $\Theta(x)$  is the Heaviside step function. The coefficients  $A, B, C, D, E, F$  describe the photonic field inside the coupling regions, and  $t_j, r_j$  are the transmission and reflection amplitudes on waveguide  $j$ . Integrating the Schrödinger equation  $H|\Psi\rangle = E|\Psi\rangle$  across the coupling points yields 12 jump conditions relating these 12 waveguide variables. Two additional atomic self-consistency equations determine the atomic excitation amplitudes  $f_a, f_b$ . Together these equations form a closed  $14 \times 14$  linear system,

$$\mathcal{A}_1 \mathbf{x}^{(1)} = \mathbf{b}_1, \quad \mathbf{x}^{(1)} = (A, B, C, D, E, F, r_1, t_1, r_2, t_2, r_3, t_3, \eta_a^{(1)}, \eta_b^{(1)})^T, \quad (8)$$

where the source vector  $\mathbf{b}_1$  contains the unit incident field on waveguide 1. The full set of 14 equations is given in Appendix B; solving this system analytically gives the six scattering amplitudes below.

### D. Factorized scattering amplitudes

Solving the  $14 \times 14$  system yields the common denominator

$$\begin{aligned} M = & -2\Gamma^2 \cos(\theta_{a2} - \theta_{b2}) - 4\Gamma^2 e^{2i\varphi_2} \sin^2 \frac{\theta_{a2} - \theta_{b2}}{2} \\ & + \tau_{a1}\tau_{b2} + (\tau_{a1} + \tau_{a2})\tau_{b3} \\ & + (2\Gamma - i\Delta)(\tau_{a2} + \tau_{b2}) + (4\Gamma - i\Delta)(\tau_{a1} + \tau_{b3}) \\ & + 14\Gamma^2 - 8i\Gamma\Delta - \Delta^2, \end{aligned} \quad (9)$$

and the six scattering amplitudes. It is useful to first express the amplitudes through the atomic excitation am-

plitudes  $\eta_a^{(1)}, \eta_b^{(1)}$  obtained from the  $14 \times 14$  system:

$$r_1 = -i\eta_a^{(1)}P_{1-}, \quad (10a)$$

$$t_1 = 1 - i\eta_a^{(1)}P_{1+}^*, \quad (10b)$$

$$r_2 = -i(\eta_a^{(1)}P_{2-}^a + \eta_b^{(1)}P_{2-}^b), \quad (10c)$$

$$t_2 = -i(\eta_a^{(1)}P_{2+}^{a*} + \eta_b^{(1)}P_{2+}^{b*}), \quad (10d)$$

$$r_3 = -i\eta_b^{(1)}P_{3-}, \quad (10e)$$

$$t_3 = -i\eta_b^{(1)}P_{3+}^*. \quad (10f)$$

Here we have introduced a systematic phase-factor notation to distinguish output directions:

$$P_{j,m}^L = 1 + e^{i(\varphi_j - \theta_{mj})}, \quad P_{j,m}^R = 1 + e^{-i(\varphi_j + \theta_{mj})}, \quad (11)$$

which for waveguides 1 and 3 specialize to  $P_{1-} \equiv P_{1,a}^L$ ,  $P_{1+}^* \equiv P_{1,a}^R$ ,  $P_{3-} \equiv P_{3,b}^L$ ,  $P_{3+}^* \equiv P_{3,b}^R$ . The ballistic “1” in  $t_1$  represents the transmitted photon that passes through without atomic interaction.

The atomic excitation amplitudes are obtained from the reduced  $2 \times 2$  atomic subsystem:

$$\eta_a^{(1)} = -i\Gamma P_{1+}(\tau_{b2} + \tau_{b3} + 4\Gamma - i\Delta)/M, \quad (12a)$$

$$\eta_b^{(1)} = \frac{-i\Gamma P_{1+}}{M P_{2-}^b} [A_r - (\tau_{b2} + \tau_{b3} + 4\Gamma - i\Delta)P_{2-}^a], \quad (12b)$$

with the full derivation provided in Appendix B. For reference, the  $\eta$ -form of Eqs. (10) reproduces the exhaustive rational expressions obtained by directly eliminating  $\eta_a^{(1)}, \eta_b^{(1)}$ . After elimination,  $r_1, t_1$ , and  $r_2$  recover the compact factorized forms

$$t_1 = N_{t1}/M, \quad (13a)$$

$$r_1 = -\Gamma P_{1-}P_{1+}(\tau_{b2} + \tau_{b3} + 4\Gamma - i\Delta)/M, \quad (13b)$$

$$r_2 = -\Gamma P_{1+}A_r/M, \quad (13c)$$

where  $N_{t1}$  is given in Appendix A. The remaining amplitudes  $t_2, t_3, r_3$  do not admit comparably simple elimination and are most compactly represented in the  $\eta$ -form of Eqs. (10d)–(10f). The elementary phase factors entering Eqs. (10) are

$$P_{1+} = 1 + e^{i(\varphi_1 + \theta_{a1})}, \quad P_{1-} = 1 + e^{i(\varphi_1 - \theta_{a1})}, \quad (14a)$$

$$P_{3+} = 1 + e^{i(\varphi_3 + \theta_{b3})}, \quad P_{3-} = 1 + e^{i(\varphi_3 - \theta_{b3})}. \quad (14b)$$

Two composite interference factors control waveguide 2:

$$\begin{aligned} A_t = & (1 + e^{i(\varphi_2 + \theta_{a2})})(\tau_{b3} + 2\Gamma - i\Delta) \\ & + \Gamma(1 - e^{2i\varphi_2})(1 - e^{i(\theta_{a2} - \theta_{b2})}), \end{aligned} \quad (15a)$$

$$\begin{aligned} A_r = & (1 + e^{i(\varphi_2 - \theta_{a2})})(\tau_{b3} + 2\Gamma - i\Delta) \\ & + \Gamma(1 - e^{2i\varphi_2})(1 - e^{i(\theta_{b2} - \theta_{a2})}). \end{aligned} \quad (15b)$$

The bridge factor coupling waveguides 2 and 3 is

$$\mathcal{B}_3 = 1 + e^{i(\theta_{a2} - \theta_{b2})} + e^{i(\varphi_2 + \theta_{a2})} + e^{i(\varphi_2 - \theta_{b2})}. \quad (16)$$

Equations (13) reveal a factorized structure for the three amplitudes that simplify under  $\eta$ -elimination:  $r_1$  contains the product  $P_{1-}P_{1+}$ ,  $r_2$  contains  $P_{1+}A_r$ , and  $t_1$  requires the full numerator  $N_{t_1}$ . The remaining amplitudes  $t_2, t_3, r_3$  are given in compact  $\eta$ -form by Eqs. (10d)–(10f), where the algebraic complexity resides in  $\eta_a^{(1)}, \eta_b^{(1)}$ .

### E. Zero-condition covering

The  $\eta$ -form of Eqs. (10) reveals that both atomic excitation amplitudes share the entrance factor  $P_{1+}$  [see Eqs. (12)]. Consequently  $P_{1+} = 0$  simultaneously kills five amplitudes ( $r_1, r_2, t_2, r_3, t_3$ ), leaving only  $t_1$  active. Beyond this global cover, each amplitude possesses individual zero conditions:

$$Z(r_1) : P_{1-} = 0 \text{ or } P_{1+} = 0, \quad (17)$$

$$Z(t_2) : \eta_a^{(1)} P_{2+}^{a*} + \eta_b^{(1)} P_{2+}^{b*} = 0, \quad (18)$$

$$Z(r_2) : \eta_a^{(1)} P_{2-}^{a*} + \eta_b^{(1)} P_{2-}^{b*} = 0, \quad (19)$$

$$Z(t_3) : P_{1+} = 0 \text{ or } P_{3+}^* = 0, \quad (20)$$

$$Z(r_3) : P_{1+} = 0 \text{ or } P_{3-} = 0. \quad (21)$$

The elementary phase-factor conditions are

$$P_{1+} = 0 \iff \varphi_1 + \theta_{a1} = (2n+1)\pi, \quad (22a)$$

$$P_{1-} = 0 \iff \varphi_1 - \theta_{a1} = (2n+1)\pi, \quad (22b)$$

$$P_{3+}^* = 0 \iff \varphi_3 + \theta_{b3} = (2n+1)\pi, \quad (22c)$$

$$P_{3-} = 0 \iff \varphi_3 - \theta_{b3} = (2n+1)\pi. \quad (22d)$$

The conditions for  $t_2$  and  $r_2$  are linear combinations of  $\eta_a^{(1)}$  and  $\eta_b^{(1)}$  and do not factorize into single elementary zeros; they must be analyzed through the full atomic subsystem (see Appendix B).

Perfect routing to a target port requires selecting zero conditions that simultaneously suppress all five non-target ports while keeping the target amplitude and  $M$  finite. Table I summarizes the representative covering strategies.

TABLE I. Representative sufficient zero-condition coverings for waveguide-1 edge incidence. Conditions  $C_2^t$  and  $C_2^r$  denote the linear-combination zeros  $\eta_a^{(1)} P_{2+}^{a*} + \eta_b^{(1)} P_{2+}^{b*} = 0$  and  $\eta_a^{(1)} P_{2-}^{a*} + \eta_b^{(1)} P_{2-}^{b*} = 0$ , respectively. The table gives constructive coverings used in this work and is not a necessary-and-sufficient classification of all mathematical solutions.

| Target | Required non-target zeros                                                                             | Target-preserving conditions               | Status                             |
|--------|-------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------|
| $T_1$  | $P_{1+} = 0$                                                                                          | $M \neq 0$                                 | Exact sufficient route             |
| $R_1$  | $N_{t_1} = 0, C_2^t = 0, C_2^r = 0, r_3 = t_3 = 0$                                                    | $P_{1+}P_{1-} \neq 0, M \neq 0$            | Composite route                    |
| $T_2$  | $N_{t_1} = 0, P_{1-} = 0, C_2^t = 0, r_3 = t_3 = 0$                                                   | $C_2^r \neq 0, M \neq 0$                   | Composite route                    |
| $R_2$  | $N_{t_1} = 0, P_{1-} = 0, C_2^t = 0, r_3 = t_3 = 0$                                                   | $C_2^r \neq 0, M \neq 0$                   | Composite route                    |
| $T_3$  | $N_{t_1} = 0, P_{1+} = 0, C_2^t = 0, C_2^r = 0, P_{3-} = 0, P_{3+}^* \neq 0, P_{1+} \neq 0, M \neq 0$ | $P_{3-} \neq 0, P_{1+} \neq 0, M \neq 0$   | Near-resonant representative route |
| $R_3$  | $N_{t_1} = 0, P_{1+} = 0, C_2^t = 0, C_2^r = 0, P_{3+}^* = 0, P_{3-} \neq 0, P_{1+} \neq 0, M \neq 0$ | $P_{3+}^* \neq 0, P_{1+} \neq 0, M \neq 0$ | Near-resonant representative route |

The covering classification naturally separates the six routing channels into three mechanism classes. **Direct phase-zero route ( $T_1$ )**. The condition  $P_{1+} = 0$  simultaneously covers all five non-target ports, leaving only the self-transmission  $t_1$  active. Provided  $M \neq 0$ , probabilistic conservation guarantees  $T_1 = 1$ . This is the cleanest

mechanism: a single phase condition on  $\varphi_1 + \theta_{a1}$  achieves perfect routing.

**Composite-interference routes ( $R_1, T_2, R_2$ )**. These channels require multiple simultaneous conditions—suppressing  $t_1$  through its numerator, closing the waveguide-2 outputs through the linear-combination conditions  $C_2^t$  and/or  $C_2^r$ , and suppressing the far waveguide by driving  $\eta_b^{(1)} \rightarrow 0$ . The number of constraints approaches the number of available parameters, making these routes exact or near-exact.

**Near-resonant singular routes ( $T_3, R_3$ )**. Routing to waveguide 3 is the most constrained case. It requires entrance locking via  $P_{1-} = 0$  (since  $P_{1+} = 0$  would also kill the target), waveguide-2 closure via  $C_2^t = C_2^r = 0$ , and direction selectivity on waveguide 3 through  $P_{3+}^* = 0$  or  $P_{3-} = 0$ . At strict resonance  $\Delta = 0$ , the waveguide-2 closure conditions force degeneracy between the two waveguide-3 directions; a small nonzero detuning  $\Delta$  breaks this degeneracy at linear order, enabling directional selection. The optimal detuning scale for the representative branch is  $|\Delta|/\Gamma \sim 4.6 \times 10^{-3}$ , consistent with the near-resonant character of this mechanism: a small but finite detuning is required to break the directional degeneracy, yet it must remain within the resonance window.

## III. FIXED-BACKBONE EDGE-INCIDENCE ROUTING LANDSCAPE

The zero-condition covering analysis of Sec. II C is independent of any particular parameter choice. To illustrate the routing mechanisms concretely, we fix the representative branch

$$\boxed{\Gamma = 1, \quad \theta_{a2} = \pi,} \quad (23)$$

which reduces the free parameter space to seven dimensions ( $\Delta, \theta_{a1}, \theta_{b2}, \theta_{b3}, \varphi_1, \varphi_2, \varphi_3$ ). All angular parameters are quoted in units of  $\pi$  below.

### A. Phase-space routing maps

Figure 2 displays two-dimensional slices through the seven-dimensional routing landscape around the high-transmission basins identified by the zero-condition coverings of Table I. The plotted coordinates are chosen channel by channel so that the relevant structure is visible: broad routes are shown on full or seam-centered phase windows, while the remote waveguide-3 routes require a magnified near-resonant window. All panels use the same probability color scale, so the areas of the high-fidelity regions can be compared directly. The remaining parameters are held at their optimized values listed in Appendix C.

The  $T_1$  panel is particularly instructive: the broad diagonal band of high transmission directly visualizes the

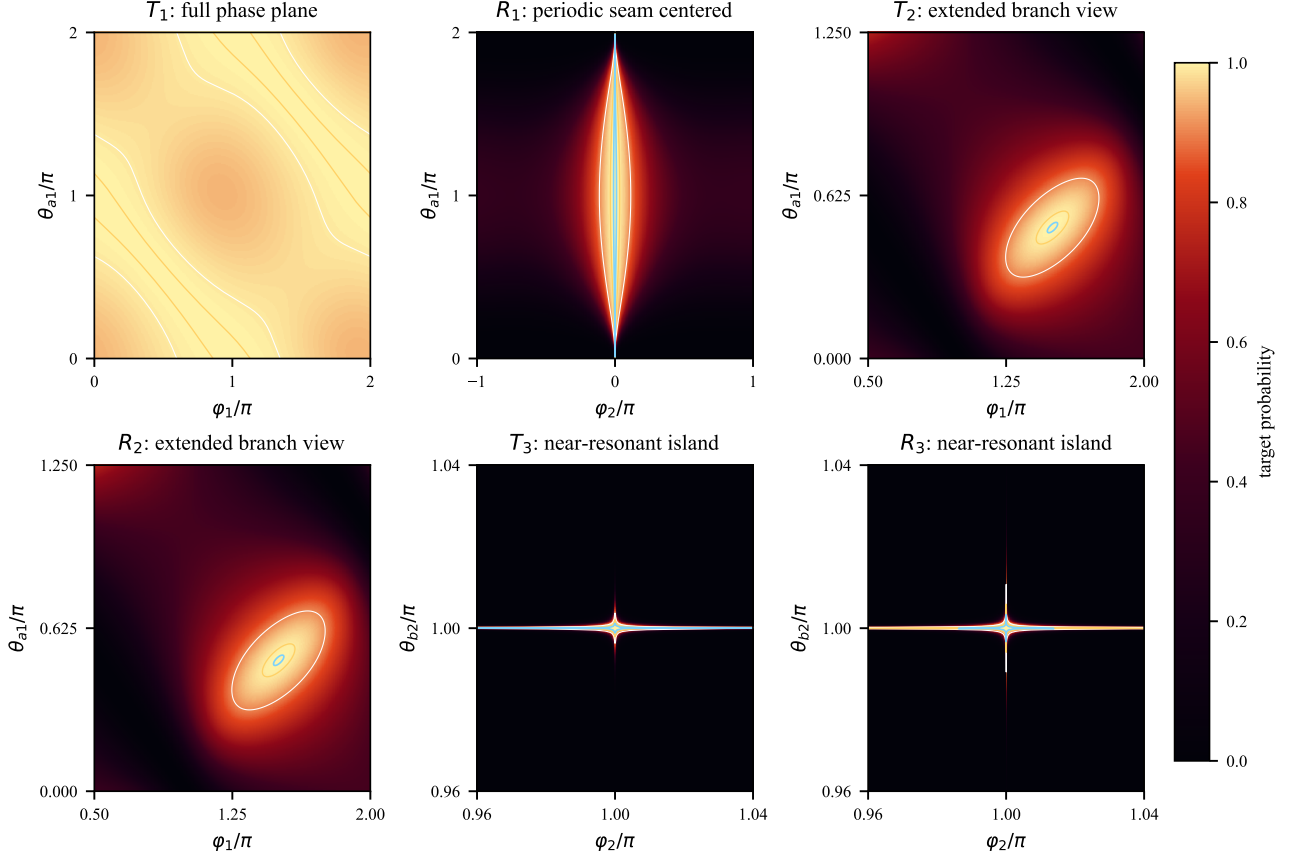


FIG. 2. Phase-coordinate routing landscapes around the six fixed-backbone optima for waveguide-1 edge incidence. The  $T_1$  panel shows the full  $0-2\pi$  phase plane, while the  $R_1$  panel is recentered at the periodic seam so that the two edge features join into one continuous high-transmission region. The  $T_2$  and  $R_2$  panels use extended branch windows large enough to show the boundary of the high-fidelity islands around the correlated branch neighborhoods introduced below. The  $T_3$  and  $R_3$  panels use magnified near-resonant windows because their high-fidelity regions are much narrower on the full phase torus. Contours mark  $P = 0.9, 0.99$ , and  $0.999$  where visible. These two-dimensional cuts demonstrate sufficient high-probability routes around representative optima; they are not intended as uniqueness proofs for the corresponding routing conditions. The exact optimized parameters are given in Table III.

sufficient direct phase-zero condition  $\varphi_1 + \theta_{a1} = (2n+1)\pi$  ( $P_{1+} = 0$ ), confirming that the entire one-parameter family of phase combinations on this line achieves near-unit routing. This condition is not claimed to be the only mathematical route to  $T_1 = 1$ ; it is the simplest constructive route because it suppresses all five non-target amplitudes at once. The width of the band reflects the insensitivity of the denominator  $M$  to the specific value of the phase sum within a reasonable tolerance.

The  $T_2$  and  $R_2$  panels exhibit compact high-probability islands, consistent with the composite-interference mechanism: these channels require the simultaneous satisfaction of multiple zero conditions, leaving a lower-dimensional solution manifold. Both are shown because the two output directions correspond to different linear-combination zeros, even though they are symmetry-related. The  $T_3$  and  $R_3$  panels show very narrow near-resonant ridges, reflecting the singular nature of the waveguide-3 routing mechanism discussed in Sec. II C.

These channels require sub-percent-level tuning of the coupling phases.

### B. Correlated branch for $T_2/R_2$ routing

The compact  $T_2$  and  $R_2$  islands in Fig. 2 become more transparent when the correlated phase branch behind the composite-interference route is imposed explicitly. For the  $T_2$  route, the optimized point lies close to the following sufficient branch. Writing phases in units of  $\pi$ , one representative branch is

$$\begin{aligned} \mathcal{C}_2(q) : \quad & \varphi_2/\pi = 1 - q, & \varphi_3/\pi = 0, & \theta_{b3}/\pi = 1, \\ & \varphi_1/\pi = \frac{3}{2} + \frac{3q}{2}, & \theta_{a1}/\pi = \frac{1}{2} + \frac{3q}{2}, \\ & \Delta/\Gamma = 2 \sin(\pi q), \end{aligned} \tag{24}$$

with  $\theta_{b2}/\pi = q$  for one  $T_2$  branch and a symmetry-related choice for  $R_2$ . This relation should be read as a constructive sufficient path through parameter space rather than as a uniqueness statement.

Figure 3 makes the role of the correlated  $T_2/R_2$  phase relations explicit. For each channel we compare two scans on the same  $(q, \eta)$  window and with the same probability color scale. In the unconstrained scan, only the local variables  $(q, \eta)$  are varied around a single high-transmission point, while the remaining phases are held fixed. In the constrained scan, the full branch  $C_2(q)$  is imposed and the same transverse deviation  $\eta$  is scanned. The high-fidelity area is then much larger, showing that the phase relations are not merely a convenient parametrization but actively enlarge the routing basin.

### C. Detuning sensitivity of near-resonant routes

The near-resonant waveguide-3 channels are also unusually sensitive to detuning. At strict resonance, the two waveguide-3 output directions are degenerate; a small but nonzero detuning breaks this degeneracy and selects one direction. Figure 4 illustrates this point for a representative  $T_3$  solution. The left panel shows that the high-fidelity window is narrow in  $\Delta$ , while the right panel compares the best target probabilities when  $\Delta$  is optimized with those obtained after forcing  $\Delta = 0$ .

### D. Small-atom reference: absence of independent directional phase zeros

To clarify the role of the giant-atom phase degrees of freedom, we briefly contrast the results above with the corresponding small-atom node. In the small-atom limit, each atom couples to its waveguides at a single point: atom  $a$  at  $x = 0$  on waveguides 1 and 2, atom  $b$  at  $x = l$  on waveguides 2 and 3. The only propagation phase is  $\phi = k_2 l = k_3 l$ , and no local phase differences  $\theta_{mj}$  exist. We use the same definitions  $\Gamma = g^2/v_g$  and  $\Delta = \omega - \omega_a = \omega - \omega_b$  as in the giant-atom model. Solving the same real-space scattering problem yields the six small-atom amplitudes. The far-waveguide outputs are particularly instructive:

$$t_3^s = \frac{\Gamma^2}{D_s}, \quad (25a)$$

$$r_3^s = \frac{e^{2i\phi}\Gamma^2}{D_s}, \quad (25b)$$

with

$$D_s = -4\Gamma^2 + e^{2i\phi}\Gamma^2 + 4i\Gamma\Delta + \Delta^2. \quad (26)$$

Consequently,

$$|t_3^s|^2 = |r_3^s|^2, \quad (27)$$

i.e., the two output directions on waveguide 3 are *probabilistically indistinguishable* within this symmetric small-atom reference model. No choice of  $\phi$  or  $\Delta$  can make  $|t_3^s|^2 \neq |r_3^s|^2$ .

In contrast, the giant-atom far-waveguide amplitudes [Eqs. (10e)–(10f)] are proportional to two *independent* phase factors,

$$t_3 \propto P_{3+}^* = 1 + e^{-i(\varphi_3 + \theta_{b3})}, \quad r_3 \propto P_{3-} = 1 + e^{i(\varphi_3 - \theta_{b3})}, \quad (28)$$

whose zeros can be controlled independently through the local phase difference  $\theta_{b3}$ . The small-atom node lacks this directional selectivity because all propagation accumulates in the single phase  $\phi$ , which factors out as a common modulus in  $|t_3^s|^2$  and  $|r_3^s|^2$ . This small-atom reference is included only to highlight the directional phase selectivity of the giant-atom node.

## IV. MIDDLE-INCIDENT SCATTERING AND THE SIX-PORT MATRIX

### A. Motivation

The edge-incidence amplitudes of Sec. II C provide one column of the  $6 \times 6$  scattering matrix (column 1 for WG1-L incidence, and by symmetry column 5 for WG3-L incidence). To complete the left-incidence description of the node we also need the response to a photon entering from the middle waveguide 2 (port 3). This middle-incidence configuration is physically distinct because the incident field drives both atoms simultaneously through their shared coupling to waveguide 2, rather than a single atom as in the edge case.

### B. Scattering ansatz for waveguide-2 incidence

For a photon incident from the left of waveguide 2, the ansatz is obtained from Eq. (7) by moving the unit incident amplitude from  $\Phi_{R,1}$  to  $\Phi_{R,2}$ :

$$\Phi_{R,1}(x) = e^{ik_1 x} [\tilde{A} \Theta(x) \Theta(l_1 - x) + \tilde{t}_1 \Theta(x - l_1)], \quad (29a)$$

$$\Phi_{L,1}(x) = e^{-ik_1 x} [\tilde{r}_1 \Theta(-x) + \tilde{B} \Theta(x) \Theta(l_1 - x)], \quad (29b)$$

$$\Phi_{R,2}(x) = e^{ik_2 x} [\Theta(-x) + \tilde{C} \Theta(x) \Theta(l_2 - x) + \tilde{t}_2 \Theta(x - l_2)], \quad (29c)$$

$$\Phi_{L,2}(x) = e^{-ik_2 x} [\tilde{r}_2 \Theta(-x) + \tilde{D} \Theta(x) \Theta(l_2 - x)], \quad (29d)$$

$$\Phi_{R,3}(x) = e^{ik_3 x} [\tilde{E} \Theta(x) \Theta(l_3 - x) + \tilde{t}_3 \Theta(x - l_3)], \quad (29e)$$

$$\Phi_{L,3}(x) = e^{-ik_3 x} [\tilde{r}_3 \Theta(-x) + \tilde{F} \Theta(x) \Theta(l_3 - x)]. \quad (29f)$$

Compared with Eq. (7), the only physical change is where the incoming unit-amplitude wave is placed. For WG1 incidence the incoming term is the “1” in  $\Phi_{R,1}$  before the first coupling point; for WG2 incidence it is the “1” in  $\Phi_{R,2}$  before the first coupling point. The Hamiltonian, coupling phases, and jump rules at the six coupling

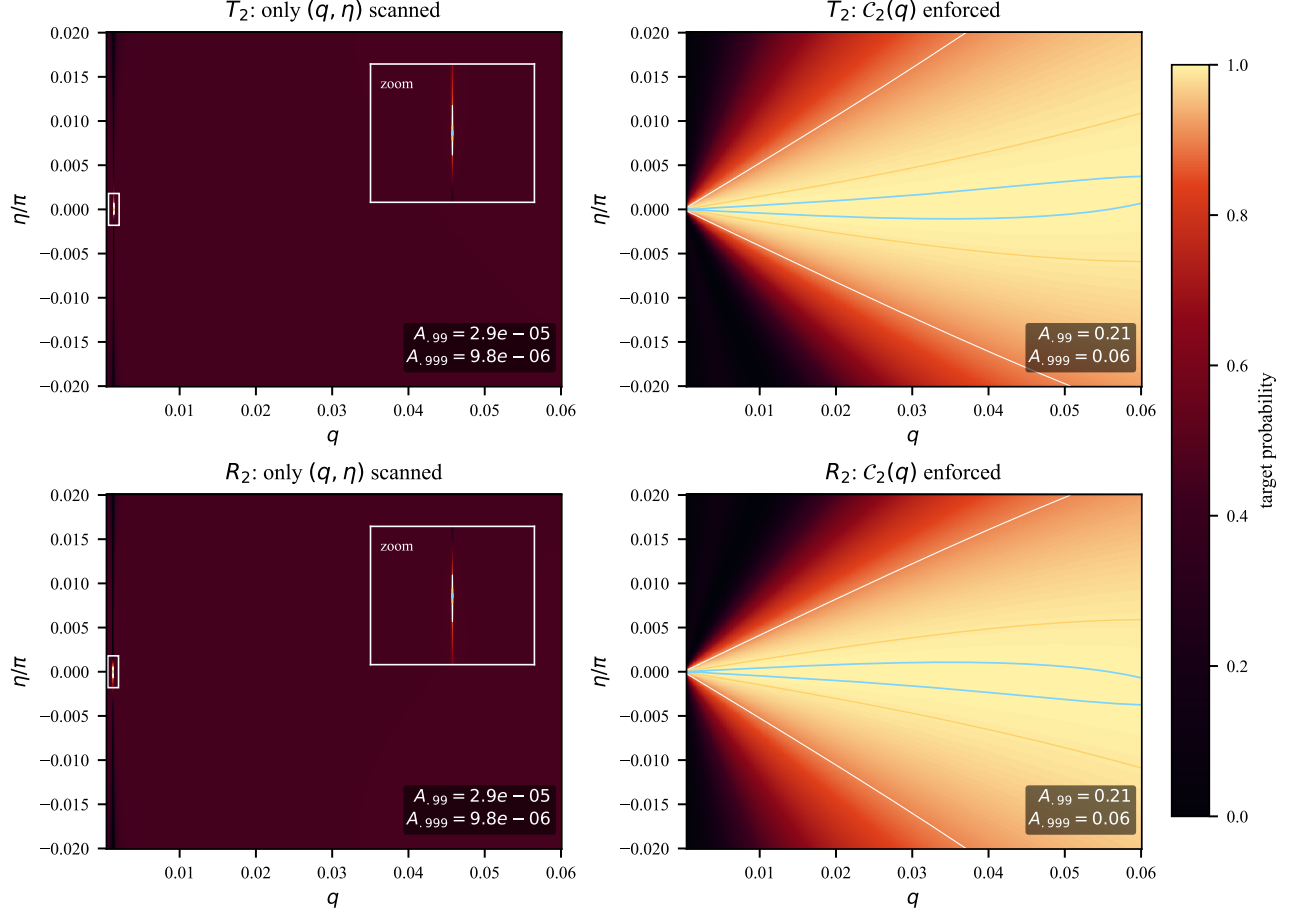


FIG. 3. Effect of enforcing the constructive  $T_2/R_2$  phase branch. Left column: only the two local variables  $(q, \eta)$  are scanned around a single high-transmission point, without enforcing the correlated relations in  $C_2(q)$ . The insets magnify the narrow high-fidelity spots in the unconstrained scans. Right column: the full  $C_2(q)$  branch is imposed and the same transverse deviation  $\eta$  is scanned. All panels use the same color scale; contours mark  $P = 0.9, 0.99$ , and  $0.999$ . The displayed area fractions  $A_{.99}$  and  $A_{.999}$  show that the derived phase constraints substantially enlarge the high-fidelity basin.

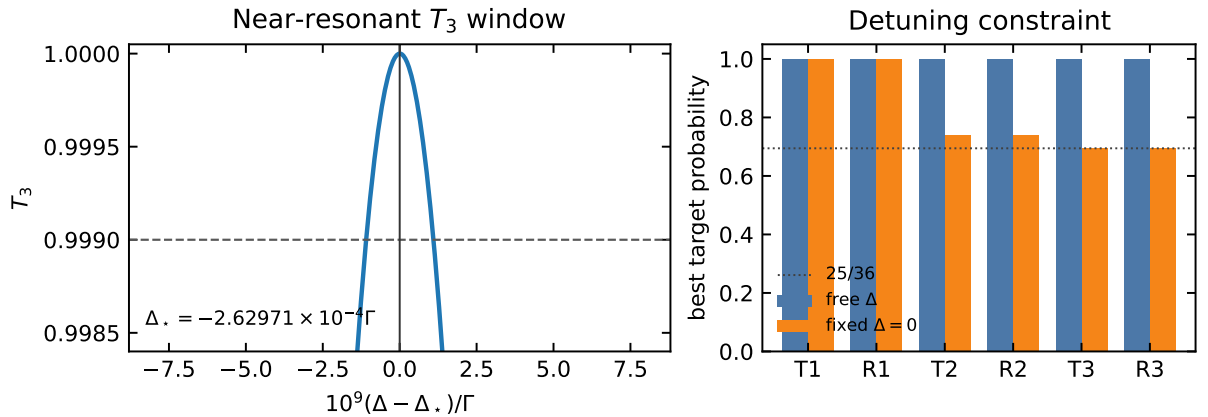


FIG. 4. Detuning sensitivity of the near-resonant waveguide-3 routing mechanism. Left: a representative  $T_3$  line shape near the optimal detuning  $\Delta_*$ ; the dashed line marks  $T_3 = 0.999$ . Right: comparison between optimizing  $\Delta$  freely and imposing strict resonance  $\Delta = 0$ . The waveguide-3 channels lose directional selectivity at strict resonance, consistent with the degeneracy argument in Sec. II C.

points are unchanged. Thus the 14 equations have the same left-hand-side coefficients as in the edge-incidence problem: the unknown internal amplitudes, output amplitudes, and atomic amplitudes enter with the same phase factors. What changes are the constant source terms generated by the incoming photon. Equivalently, in the matrix form  $\mathcal{A}\mathbf{x} = \mathbf{b}$ , the matrix  $\mathcal{A}$  is the same and the source vector  $\mathbf{b}$  is changed. This is why the middle-incidence amplitudes share the same denominator  $M$  as Eq. (9).

### C. Scattering amplitudes in compact form

Solving the 12 jump conditions for the internal waveguide amplitudes in terms of  $\eta_a^{(3)}, \eta_b^{(3)}$  (see Appendix B for details) and substituting into the remaining relations yields the six scattering amplitudes:

$$\tilde{t}_1 = -i\eta_a^{(3)} P_{1,a}^R, \quad \tilde{r}_1 = -i\eta_a^{(3)} P_{1,a}^L, \quad (30a)$$

$$\tilde{t}_2 = 1 - i\eta_a^{(3)} P_{2,a}^R - i\eta_b^{(3)} P_{2,b}^R, \quad (30b)$$

$$\tilde{r}_2 = -i\eta_a^{(3)} P_{2,a}^L - i\eta_b^{(3)} P_{2,b}^L, \quad (30c)$$

$$\tilde{t}_3 = -i\eta_b^{(3)} P_{3,b}^R, \quad \tilde{r}_3 = -i\eta_b^{(3)} P_{3,b}^L. \quad (30d)$$

Here we have introduced a systematic phase-factor notation to distinguish the three physically distinct roles played by phase factors in the scattering problem:

$$\begin{aligned} P_{j,m}^{\text{src}} &= 1 + e^{i(\varphi_j + \theta_{mj})}, \\ P_{j,m}^R &= 1 + e^{-i(\varphi_j + \theta_{mj})}, \\ P_{j,m}^L &= 1 + e^{i(\varphi_j - \theta_{mj})}, \end{aligned} \quad (31)$$

corresponding respectively to the source driving factor (coupling of the incident field to the atom), the right-moving output factor, and the left-moving output (reflection) factor. Note that  $P_{j,m}^R = (P_{j,m}^{\text{src}})^*$ , while  $P_{j,m}^L$  is an independent factor.

The term “1” in  $\tilde{t}_2$  represents the ballistic transmission background—a photon that passes through waveguide 2 without interacting with either atom. The actual value of  $\tilde{t}_2$  is modified from unity by the scattering contributions proportional to  $\eta_a^{(3)}$  and  $\eta_b^{(3)}$ .

### D. Atomic excitation amplitudes

The atomic amplitudes  $\eta_a^{(3)}, \eta_b^{(3)}$  are obtained by substituting the jump-condition relations into the atomic self-consistency equations. The  $14 \times 14$  linear system reduces to a  $2 \times 2$  system for  $(\eta_a^{(3)}, \eta_b^{(3)})$ , whose solution factorizes in terms of the *same* composite factors that appear

in the edge-incidence result:

$$\eta_a^{(3)} = -i\Gamma \frac{A_t}{M}, \quad (32a)$$

$$\eta_b^{(3)} = -i\Gamma \frac{A_t^b}{M}, \quad (32b)$$

where  $A_t$  is given by Eq. (15) and  $A_t^b$  is its  $a \leftrightarrow b$  symmetric counterpart,

$$\begin{aligned} A_t^b &= (1 + e^{i(\varphi_2 + \theta_{b2})})(\tau_{a1} + 2\Gamma - i\Delta) + \Gamma(1 - e^{2i\varphi_2}) \\ &\quad \times (1 - e^{i(\theta_{b2} - \theta_{a2})}). \end{aligned} \quad (33)$$

The complete derivation, including the  $2 \times 2$  reduction and the verification by back-substitution into all 14 Schrödinger equations, is presented in Appendix B.

Substituting Eqs. (32) into Eqs. (30) gives the waveguide-2 scattering amplitudes in fully explicit rational form:

$$\tilde{t}_1 = -\Gamma P_{1,a}^R A_t / M, \quad \tilde{r}_1 = -\Gamma P_{1,a}^L A_t / M, \quad (34a)$$

$$\tilde{t}_2 = 1 - \Gamma (P_{2,a}^R A_t + P_{2,b}^R A_t^b) / M, \quad (34b)$$

$$\tilde{r}_2 = -\Gamma (P_{2,a}^L A_t + P_{2,b}^L A_t^b) / M, \quad (34c)$$

$$\tilde{t}_3 = -\Gamma P_{3,b}^R A_t^b / M, \quad \tilde{r}_3 = -\Gamma P_{3,b}^L A_t^b / M. \quad (34d)$$

Equations (30)–(34) should be read in two steps. First, once the atomic excitation amplitudes  $\eta_a^{(3)}$  and  $\eta_b^{(3)}$  are known, each output amplitude is obtained by multiplying the relevant atomic amplitude by the phase factor for that output direction. For example, emission from atom  $a$  into the right side of waveguide 1 gives  $\tilde{t}_1 \propto \eta_a^{(3)} P_{1,a}^R$ , while emission into the left side gives  $\tilde{r}_1 \propto \eta_a^{(3)} P_{1,a}^L$ . The middle-waveguide outputs contain contributions from both atoms because both atoms couple to waveguide 2. Second, the atomic amplitudes themselves are driven by the incident photon through the two composite numerators  $A_t$  and  $A_t^b$ . Here  $A_t$  is the effective drive of atom  $a$  after all interference paths through waveguide 2 are included,  $A_t^b$  is the corresponding drive of atom  $b$ , and  $M$  is the common response denominator of the coupled two-atom system. Thus the formula has a simple physical structure: source interference determines  $\eta_a^{(3)}$  and  $\eta_b^{(3)}$ , and these atomic excitations then radiate into the six output ports with direction-dependent phase factors.

### E. Construction of the complete $6 \times 6$ scattering matrix

The three independent left-incidence column vectors, ordered according to the port convention of Eq. (1), are

$$\begin{aligned} \mathbf{s}^{(1)} &= (r_1, t_1, r_2, t_2, r_3, t_3)^\top, \\ \mathbf{s}^{(3)} &= (\tilde{r}_1, \tilde{t}_1, \tilde{r}_2, \tilde{t}_2, \tilde{r}_3, \tilde{t}_3)^\top, \\ \mathbf{s}^{(5)} &= (\hat{r}_1, \hat{t}_1, \hat{r}_2, \hat{t}_2, \hat{r}_3, \hat{t}_3)^\top, \end{aligned} \quad (35)$$



corresponding to incidence from the left side of waveguides 1, 2, and 3, respectively. Column 1 is given by Eqs. (10) together with Eqs. (12), column 3 by Eqs. (30) together with Eqs. (32), and column 5 follows from column 1 by the  $a \leftrightarrow b$  atom-exchange symmetry:

$$\begin{aligned} \theta_{a1} &\leftrightarrow \theta_{b3}, & \theta_{a2} &\leftrightarrow -\theta_{b2}, & \theta_{b2} &\leftrightarrow -\theta_{a2}, \\ \theta_{b3} &\leftrightarrow \theta_{a1}, & \varphi_1 &\leftrightarrow \varphi_3, \end{aligned} \quad (36)$$

followed by remapping the output port order  $(r_1, t_1, r_2, t_2, r_3, t_3) \rightarrow (\hat{r}_3, \hat{t}_3, \hat{r}_2, \hat{t}_2, \hat{r}_1, \hat{t}_1)$ .

The remaining three right-incidence columns  $\mathbf{s}^{(2)}, \mathbf{s}^{(4)}, \mathbf{s}^{(6)}$  are obtained by independently solving the same  $14 \times 14$  linear system with the appropriate incident boundary conditions (a left-moving photon entering from the right side of each waveguide). The  $\eta$  form for these columns is constructed analogously to Eqs. (10) and (30); the complete set of machine-readable  $\eta$ -form expressions for all six ports is included in the online repository prepared for this work.

Importantly, the system is *not* reciprocal in the sense  $S = S^\top$  when local coupling phases are nonzero. The phase-reversed configuration is the time-reversal (TR) partner of the original system, which for the scattering matrix implies the probability-level partner relation

$$T_{ij}(\theta_{a1}, \theta_{a2}, \theta_{b2}, \theta_{b3}) = T_{ji}(-\theta_{a1}, -\theta_{a2}, -\theta_{b2}, -\theta_{b3}), \quad (37)$$

where  $T_{ij}(\theta) \equiv |S_{ij}(\theta)|^2$ . This relation is verified to machine precision on 5000 random parameter points, with maximum deviation below  $10^{-15}$ . At zero local phases ( $\theta_{a1} = \theta_{a2} = \theta_{b2} = \theta_{b3} = 0$ ), the standard reciprocity  $T_{ij} = T_{ji}$  is recovered.

The full  $6 \times 6$  scattering matrix is constructed from the six independently solved columns:

$$S_{\text{dir}} = (\mathbf{s}^{(1)} \ \mathbf{s}^{(2)} \ \mathbf{s}^{(3)} \ \mathbf{s}^{(4)} \ \mathbf{s}^{(5)} \ \mathbf{s}^{(6)}) = \mathcal{P} \mathcal{A}^{-1} \mathcal{B}, \quad (38)$$

where  $\mathcal{A}$  is the  $14 \times 14$  coefficient matrix [Eq. (9) gives  $-\det \mathcal{A}$ ],  $\mathcal{B}$  encodes the six incident boundary conditions, and  $\mathcal{P}$  projects onto the six output ports. Each column satisfies probability conservation to machine precision ( $\max_j |\sum_i |S_{ij}|^2 - 1| < 3 \times 10^{-15}$ ) and the full matrix is unitary ( $\|S^\dagger S - I\|_\infty < 3 \times 10^{-15}$ ). Explicit rational expressions for selected matrix elements are given above and in Appendix B; the complete  $6 \times 6$  scattering matrix is available through the machine-readable direct-solver implementation in the online repository prepared for this work.

## F. Routing probability landscape

For routing applications, the quantities of physical interest are the probabilities  $|S_{ij}|^2$ . Under the fixed backbone  $\Gamma = 1, \theta_{a2} = \pi$ , the phase-factor magnitudes reduce

to convenient trigonometric forms:

$$|P_{j,m}^{\text{src}}| = |P_{j,m}^R| = 2 \left| \cos \frac{\varphi_j + \theta_{mj}}{2} \right|, \quad (39)$$

$$|P_{j,m}^L| = 2 \left| \cos \frac{\varphi_j - \theta_{mj}}{2} \right|, \quad (40)$$

$$|\mathcal{B}_3| = 4 \left| \sin \frac{\varphi_2}{2} \sin \frac{\theta_{b2}}{2} \right|. \quad (41)$$

These trigonometric forms make the zero-condition covering physically transparent: each factor zero corresponds to a half-angle sine or cosine vanishing at a specific phase combination.

## G. Routing under middle-waveguide incidence

The middle-incidence formulas of Eqs. (30)–(32) contain an exact transparent-transmission point that can be demonstrated analytically. Under the fixed backbone  $\Gamma = 1, \theta_{a2} = \pi$ , choosing

$$\varphi_2 = 2\pi, \quad \theta_{b2} = \pi, \quad (42)$$

gives  $P_{2,a}^{\text{src}} = 1 + e^{i(\varphi_2 + \theta_{a2})} = 0$  and  $P_{2,b}^{\text{src}} = 1 + e^{i(\varphi_2 + \theta_{b2})} = 0$ . Since  $1 - e^{2i\varphi_2} = 0$  also vanishes at  $\varphi_2 = 2\pi$ , both composite factors evaluate to zero:

$$A_t = A_t^b = 0. \quad (43)$$

From Eqs. (32) we then have  $\eta_a^{(3)} = \eta_b^{(3)} = 0$ : the incident field drives neither atomic excitation. Substituting into Eqs. (30) yields

$$\tilde{t}_2 = 1, \quad \tilde{r}_1 = \tilde{t}_1 = \tilde{r}_2 = \tilde{r}_3 = \tilde{t}_3 = 0, \quad (44)$$

i.e., perfect ballistic transmission through the middle waveguide with all other output ports dark. This result confirms that the same factorized algebraic framework that governs edge-incidence routing also describes exact routing conditions for middle-waveguide incidence, without requiring numerical optimization. A systematic search for the remaining five middle-incidence target outputs is not pursued here; those routes are naturally handled by the same direct-oracle framework and are left for the repository-level numerical exploration.

## V. DISCUSSION

The propagation phases  $\varphi_j = k_j l_j$  have been treated as frequency-independent static parameters throughout this work, corresponding to the Markovian limit in which the photon propagation time between coupling points is negligible compared with the atomic lifetime. For finite waveguide lengths, the wave number acquires a detuning dependence  $\varphi_j(\Delta) = \phi_j + \mu_j \Delta$  with  $\mu_j = l_j/v_g$ , converting the static routing conditions into frequency-dependent interference filters. For the direct phase-zero  $T_1$  route, the

condition  $P_{1+} = 0$  is restored at periodically spaced detunings  $\Delta_n = 2n\pi/\mu_1$ , yielding a frequency-comb routing pattern. A systematic investigation of finite-delay effects on the composite-interference and near-resonant channels, including scattering-pole shifts, Wigner delays, and the transition from Markovian to non-Markovian line shapes, is left for future work.

On the experimental side, giant atoms with multiple coupling points have been demonstrated in superconducting circuit-QED architectures [6], where the coupling points are realized as capacitively coupled microwave resonators or direct galvanic contacts to a transmission line. The local coupling phases  $\theta_{mj}$  appearing in our model are *synthetic* coupling phases, distinct from the propagation phases  $\varphi_j$ . They represent controllable complex coupling phases of the type used in recent phase-controlled giant-molecule scattering theory [8]. In superconducting-circuit implementations, analogous phase or sign control could be pursued with flux-tunable SQUID-mediated couplings or other complex coupling paths, although the detailed circuit design is beyond the scope of the present work. The fixed backbone  $\theta_{a2} = \pi$  corresponds to a  $\pi$ -shifted second coupling point for atom  $a$  on waveguide 2; at the circuit level this would amount to implementing an effective negative coupling on that branch. We emphasize that the present work is a theoretical proposal; the complete six-port node structure has not yet been demonstrated experimentally, and practical implementations will require addressing finite-delay corrections, waveguide dispersion, and fabrication tolerances.

## VI. CONCLUSION

We have presented an analytical derivation of the scattering amplitudes for a two-giant-atom three-waveguide six-port node. The key structural insight is the  $\eta$ -form representation: all six output amplitudes for any incident port are expressed through the atomic excitation amplitudes  $\eta_a^{(j)}, \eta_b^{(j)}$  and a small set of elementary phase factors. For waveguide-1 edge incidence, the amplitudes  $r_1, t_1$ , and  $r_2$  admit compact factorized rational forms;  $t_2, t_3$ , and  $r_3$  are most compactly represented in  $\eta$ -form. The zero-condition covering analysis identifies three representative routing mechanism classes. The middle-waveguide incidence amplitudes and the right-incidence columns share the same  $\eta$ -form structure. By independently

solving the  $14 \times 14$  linear system for all six incident ports, we construct the complete  $6 \times 6$  scattering matrix  $S_{\text{dir}} = \mathcal{P}\mathcal{A}^{-1}\mathcal{B}$ , which is unitary and satisfies the time-reversal partner relation  $T_{ij}(\theta) = T_{ji}(-\theta)$ . Under the representative fixed backbone  $\Gamma = 1, \theta_{a2} = \pi$ , the analytical zero-condition coverings correspond to extended high-probability basins in the seven-dimensional phase space, demonstrating that selected routing channels can be addressed with near-unit probability. The extension to the non-Markovian regime and the experimental implementation of the phase-controlled routing protocols are natural directions for future investigation.

## Appendix A: The $t_1$ numerator

For completeness, we record the full expression for the  $t_1$  numerator appearing in Eq. (13a):

$$\begin{aligned} N_{t_1} = & -2\Gamma^2 \cos(\theta_{a2} - \theta_{b2}) + 2i\Gamma e^{-i\theta_{a1}} \sin \varphi_1 \\ & \times (\tau_{b2} + \tau_{b3} + 4\Gamma - i\Delta) \\ & - 4\Gamma^2 e^{2i\varphi_2} \sin^2 \frac{\theta_{a2} - \theta_{b2}}{2} + \tau_{a2}\tau_{b3} \\ & - i\Delta\tau_{b2} \\ & + (2\Gamma - i\Delta)(\tau_{a2} + \tau_{b3}) \\ & + (6\Gamma^2 - 6i\Gamma\Delta - \Delta^2). \end{aligned} \quad (\text{A1})$$

## Appendix B: Derivation of WG2 scattering amplitudes and the $2 \times 2$ atomic reduction

This appendix records the explicit linear systems used in the derivation. The waveguide-1 system shows how the  $14 \times 14$  problem is assembled from jump conditions and atomic self-consistency equations; the waveguide-2 reduction then illustrates how the same machinery gives the middle-incidence amplitudes.

### 1. Complete $14 \times 14$ system for waveguide-1 incidence

For the edge-incidence ansatz of Eq. (7), the 12 jump conditions in the gauge  $\theta_{j,1}^m = 0$  are

$$-i(A-1) + \eta_a^{(1)} = 0, \quad (\text{B1a})$$

$$i(B-r_1) + \eta_a^{(1)} = 0, \quad (\text{B1b})$$

$$-ie^{i\varphi_1}(t_1-A) + \eta_a^{(1)}e^{-i\theta_{a1}} = 0, \quad (\text{B1c})$$

$$-ie^{-i\varphi_1}B + \eta_a^{(1)}e^{-i\theta_{a1}} = 0, \quad (\text{B1d})$$

$$-iC + \eta_a^{(1)} + \eta_b^{(1)} = 0, \quad (\text{B1e})$$

$$i(D-r_2) + \eta_a^{(1)} + \eta_b^{(1)} = 0, \quad (\text{B1f})$$

$$-ie^{i\varphi_2}(t_2-C) + \eta_a^{(1)}e^{-i\theta_{a2}} + \eta_b^{(1)}e^{-i\theta_{b2}} = 0, \quad (\text{B1g})$$

$$-ie^{-i\varphi_2}D + \eta_a^{(1)}e^{-i\theta_{a2}} + \eta_b^{(1)}e^{-i\theta_{b2}} = 0, \quad (\text{B1h})$$

$$-iE + \eta_b^{(1)} = 0, \quad (\text{B1i})$$

$$i(F-r_3) + \eta_b^{(1)} = 0, \quad (\text{B1j})$$

$$-ie^{i\varphi_3}(t_3-E) + \eta_b^{(1)}e^{-i\theta_{b3}} = 0, \quad (\text{B1k})$$

$$-ie^{-i\varphi_3}F + \eta_b^{(1)}e^{-i\theta_{b3}} = 0. \quad (\text{B1l})$$

The remaining two equations are obtained by projecting the Schrödinger equation onto the two atomic excited states. With the discontinuous photonic fields evaluated by the symmetric average at each coupling point, they read

$$\begin{aligned} \frac{\Delta}{\Gamma}\eta_a^{(1)} = \frac{1}{2} & \left[ 1 + A + r_1 + B + C + r_2 + D \right. \\ & + e^{i\theta_{a1}}(e^{i\varphi_1}(A+t_1) + e^{-i\varphi_1}B) \\ & \left. + e^{i\theta_{a2}}(e^{i\varphi_2}(C+t_2) + e^{-i\varphi_2}D) \right], \end{aligned} \quad (\text{B1m})$$

$$\begin{aligned} \frac{\Delta}{\Gamma}\eta_b^{(1)} = \frac{1}{2} & \left[ C + r_2 + D + E + r_3 + F \right. \\ & + e^{i\theta_{b2}}(e^{i\varphi_2}(C+t_2) + e^{-i\varphi_2}D) \\ & \left. + e^{i\theta_{b3}}(e^{i\varphi_3}(E+t_3) + e^{-i\varphi_3}F) \right]. \end{aligned} \quad (\text{B1n})$$

Equations (B1) are the explicit component form of Eq. (8). Eliminating  $A, \dots, F, r_j, t_j$  from this system gives the reduced atomic amplitudes in Eqs. (12); substituting them back yields the compact  $\eta$ -form amplitudes in Eqs. (10).

## 2. Jump conditions in the $\theta_{j,1}^m = 0$ gauge

For waveguide-2 left incidence ( $\text{src}_1 = 0, \text{src}_3 = 1$ ), the 12 jump conditions are

$$\begin{aligned} \text{WG1: } & -i(\tilde{A}-0) + \eta_a^{(3)} = 0, \quad i(\tilde{B}-\tilde{r}_1) + \eta_a^{(3)} = 0, \\ & -ie^{i\varphi_1}(\tilde{t}_1-\tilde{A}) + \eta_a^{(3)}e^{-i\theta_{a1}} = 0, \\ & -ie^{-i\varphi_1}\tilde{B} + \eta_a^{(3)}e^{-i\theta_{a1}} = 0, \end{aligned} \quad (\text{B2})$$

$$\begin{aligned} \text{WG2: } & -i(\tilde{C}-1) + \eta_a^{(3)} + \eta_b^{(3)} = 0, \quad i(\tilde{D}-\tilde{r}_2) + \eta_a^{(3)} + \eta_b^{(3)} = 0, \\ & -ie^{i\varphi_2}(\tilde{t}_2-\tilde{C}) + \eta_a^{(3)}e^{-i\theta_{a2}} + \eta_b^{(3)}e^{-i\theta_{b2}} = 0, \\ & -ie^{-i\varphi_2}\tilde{D} + \eta_a^{(3)}e^{-i\theta_{a2}} + \eta_b^{(3)}e^{-i\theta_{b2}} = 0, \end{aligned} \quad (\text{B3})$$

$$\begin{aligned} \text{WG3: } & -i\tilde{E} + \eta_b^{(3)} = 0, \quad i(\tilde{F}-\tilde{r}_3) + \eta_b^{(3)} = 0, \\ & -ie^{i\varphi_3}(\tilde{t}_3-\tilde{E}) + \eta_b^{(3)}e^{-i\theta_{b3}} = 0, \\ & -ie^{-i\varphi_3}\tilde{F} + \eta_b^{(3)}e^{-i\theta_{b3}} = 0. \end{aligned} \quad (\text{B4})$$

### 3. Internal amplitudes

Solving the six equations that do not involve scattering amplitudes gives

$$\begin{aligned}\tilde{A} &= -i\eta_a^{(3)}, & \tilde{B} &= -i\eta_a^{(3)}e^{i(\varphi_1-\theta_{a1})}, \\ \tilde{C} &= 1 - i\eta_a^{(3)} - i\eta_b^{(3)}, \\ \tilde{D} &= -i\eta_a^{(3)}e^{i(\varphi_2-\theta_{a2})} - i\eta_b^{(3)}e^{i(\varphi_2-\theta_{b2})}, \\ \tilde{E} &= -i\eta_b^{(3)}, & \tilde{F} &= -i\eta_b^{(3)}e^{i(\varphi_3-\theta_{b3})}.\end{aligned}\quad (\text{B5})$$

Substituting these into the remaining six jump conditions yields Eqs. (30).

### 4. Reduction to a $2 \times 2$ system

Substituting the internal and scattering amplitudes into the atomic self-consistency equations produces a linear  $2 \times 2$  system for  $(\eta_a^{(3)}, \eta_b^{(3)})$ :

$$\begin{pmatrix} \Gamma\alpha_{aa} - \Delta & \Gamma\alpha_{ab} \\ \Gamma\alpha_{ba} & \Gamma\alpha_{bb} - \Delta \end{pmatrix} \begin{pmatrix} \eta_a^{(3)} \\ \eta_b^{(3)} \end{pmatrix} = -\Gamma \begin{pmatrix} \mathcal{S}_a^{\text{free}} \\ \mathcal{S}_b^{\text{free}} \end{pmatrix}, \quad (\text{B6})$$

where the coefficients (for the general  $\theta_{a2}$  case) are

$$\alpha_{aa} = -i(P_{1,a}^L + P_{1,a}^{\text{src}} + P_{2,a}^L + P_{2,a}^{\text{src}}), \quad (\text{B7})$$

$$\alpha_{ab} = -i(P_{2,b}^L + e^{i(\varphi_2+\theta_{a2})} + e^{i(\theta_{a2}-\theta_{b2})}), \quad (\text{B8})$$

$$\alpha_{ba} = -i(P_{2,a}^L + e^{i(\varphi_2+\theta_{b2})} + e^{i(\theta_{b2}-\theta_{a2})}), \quad (\text{B9})$$

$$\alpha_{bb} = -i(P_{2,b}^L + P_{2,b}^{\text{src}} + P_{3,b}^L + P_{3,b}^{\text{src}}), \quad (\text{B10})$$

$$\mathcal{S}_a^{\text{free}} = 1 + e^{i(\varphi_2+\theta_{a2})}, \quad \mathcal{S}_b^{\text{free}} = 1 + e^{i(\varphi_2+\theta_{b2})}. \quad (\text{B11})$$

Solving this  $2 \times 2$  system by Cramer's rule and rewriting the numerators in terms of the composite interference factors gives  $A_t$  and  $A_t^b$  in Eqs. (32) and (33).

### 5. Verification

The complete set of formulas has been verified by back-substitution into all 14 Schrödinger equations and by direct comparison with the  $14 \times 14$  numerical solver. Table II summarizes the validation results under the fixed backbone  $\Gamma = 1, \theta_{a2} = \pi$ ; extended verification at general parameter values is included in the online repository prepared for this work.

The verification code is organized in the online repository prepared at <https://github.com/KarasjokMing/paper2-results> and will be made public upon submission or publication.

### Appendix C: Optimized routing parameters

Table III lists the independently optimized parameter sets for waveguide-1 edge incidence that achieve the routing probabilities shown in Fig. 2.

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TABLE II. Verification of analytical formulas against the  $14 \times 14$  numerical solver. Each test uses 5000 random parameter points under the fixed backbone  $\Gamma = 1, \theta_{a2} = \pi$ .

| Verification item                                       | Max. error            |
|---------------------------------------------------------|-----------------------|
| WG1-L: $\eta$ -form (6 amplitudes) vs. numerical        | $< 2 \times 10^{-15}$ |
| WG2-L: $\eta$ -form (6 amplitudes) vs. numerical        | $2.5 \times 10^{-15}$ |
| All 6 ports: $\eta$ -form vs. numerical                 | $< 3 \times 10^{-15}$ |
| $\eta_a^{(1)}$ factorization vs. numerical              | $2.4 \times 10^{-15}$ |
| $\eta_a^{(3)}, \eta_b^{(3)}$ formulas vs. numerical     | $2.2 \times 10^{-15}$ |
| Probability conservation, all columns                   | $2.2 \times 10^{-15}$ |
| TR partner $  S_{ij}(\theta) ^2 -  S_{ji}(-\theta) ^2 $ | $3.5 \times 10^{-15}$ |
| 14-equation back-substitution residual                  | $1.9 \times 10^{-15}$ |

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TABLE III. Optimized parameter sets for waveguide-1 edge incidence,  $\Gamma = 1, \theta_{a2} = \pi$ . Angles normalized by  $\pi$ .

| Target | $\Delta/\Gamma$ | $\theta_{a1}/\pi$    | $\theta_{b2}/\pi$ | $\theta_{b3}/\pi$ | $\varphi_1/\pi$ | $\varphi_2/\pi$ | $\varphi_3/\pi$      |
|--------|-----------------|----------------------|-------------------|-------------------|-----------------|-----------------|----------------------|
| $T_1$  | 0               | 1.000                | 0.500             | 0.500             | 0.000           | 1.000           | 1.000                |
| $R_1$  | 0               | 1.001                | 0.417             | 0.363             | 1               | 0               | 1.909                |
| $T_2$  | 0.1211          | 0.5287               | 0.0194            | 1                 | 1.5285          | 0.9807          | 0                    |
| $R_2$  | -0.0992         | 1.4759               | 0.0158            | 1                 | 0.4760          | 1.0158          | 2                    |
| $T_3$  | -0.00464        | $7.4 \times 10^{-4}$ | 1.00004           | 1.00074           | 1.00074         | 1               | $7.4 \times 10^{-4}$ |
| $R_3$  | -0.00456        | $7.4 \times 10^{-4}$ | 0.99966           | 0.99926           | 1.00074         | 1.00014         | $7.4 \times 10^{-4}$ |

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